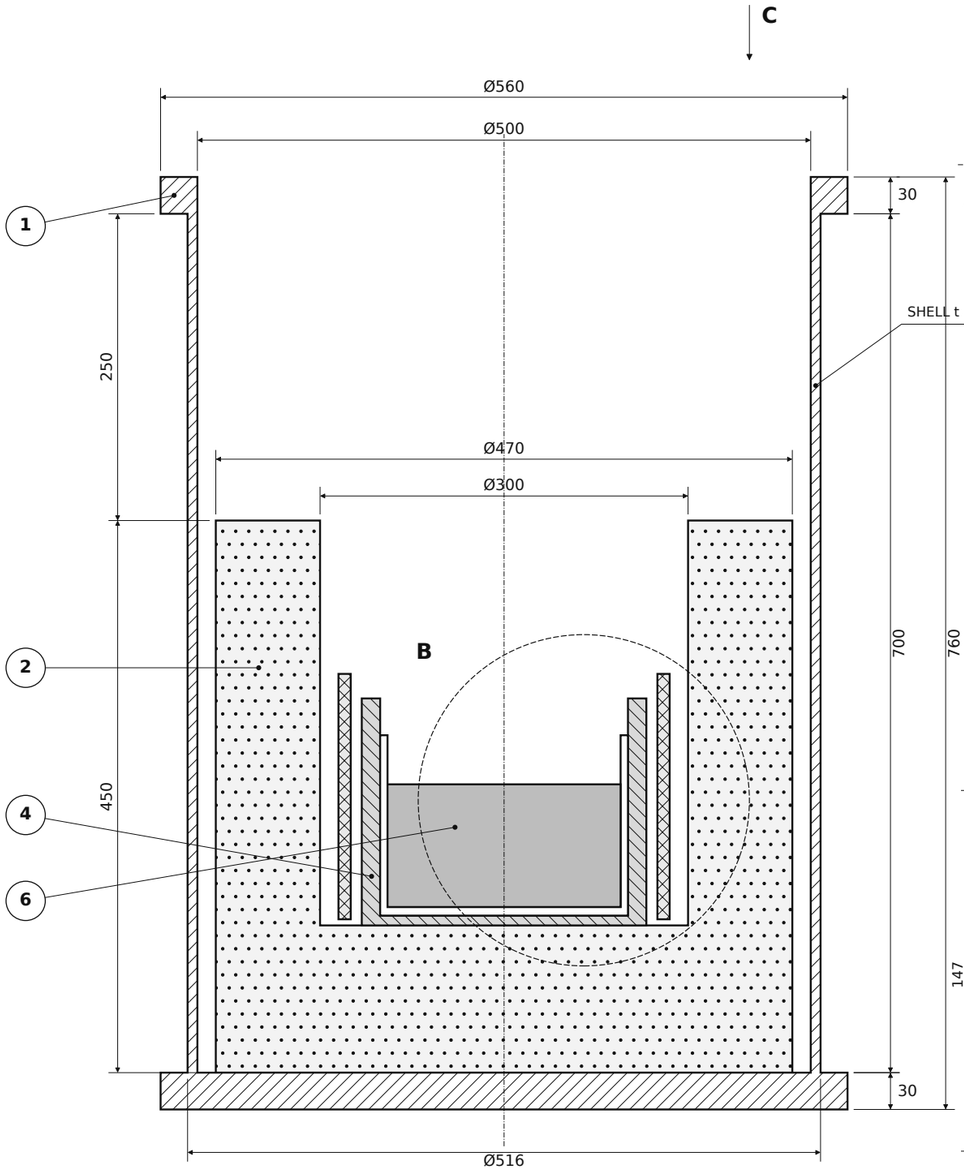
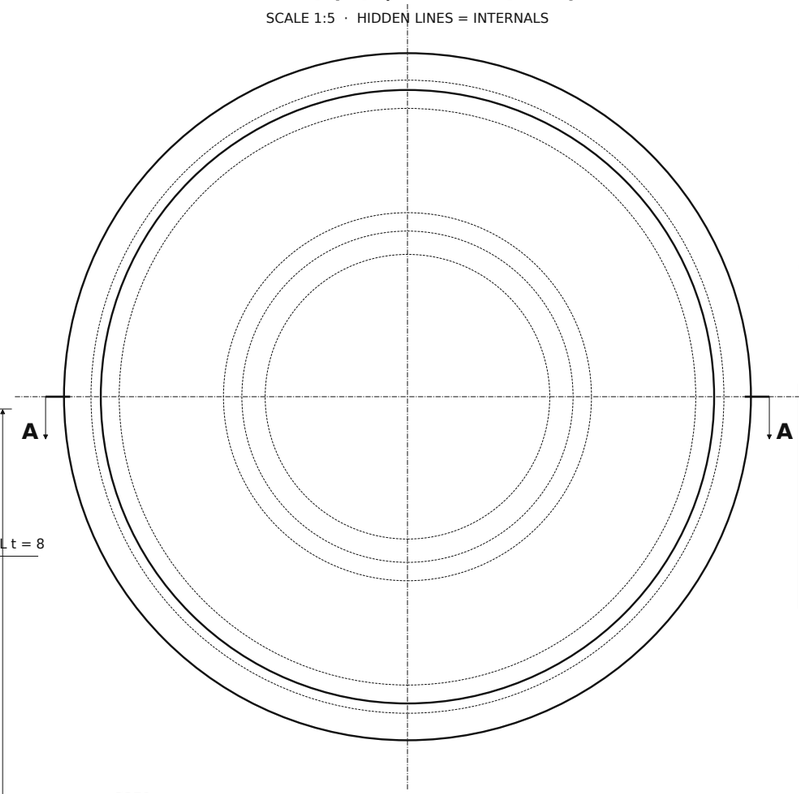


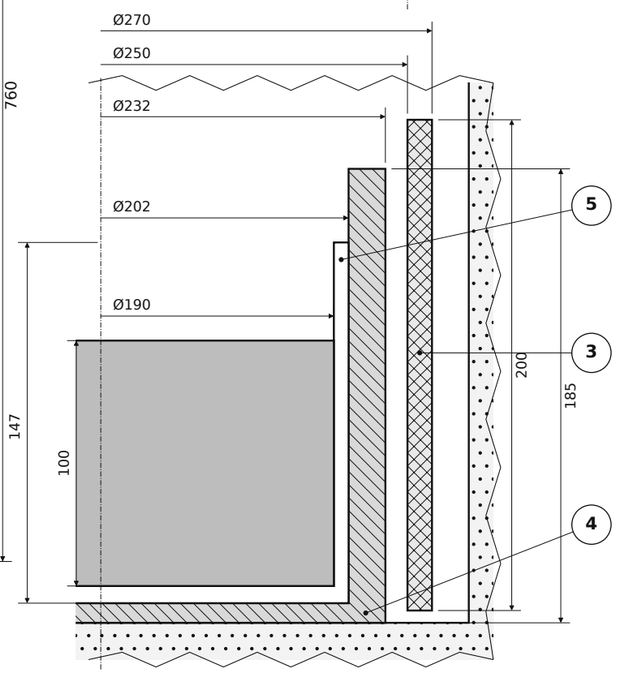

**SECTION A–A**

SCALE 1:5

## VIEW C (TOP, LID REMOVED)

SCALE 1:5 · HIDDEN LINES = INTERNALS




**DETAIL B**

SCALE 1:2.5

## GENERAL NOTES

1. All dimensions in mm. Do not scale.
2. Concept design – not for construction. Not built or experimentally validated.
3. Design basis: 1.0 MPa internal; full vacuum (0.101 MPa external). Indicative sizing to ASME VIII-1 UG-27 (shell) and UG-34 (flat head).
4. Ports (viewport, pyrometer, vacuum pump), lid, water jacket and coil leads not shown in Rev A.
5. Heater modelled as a solid graphite ring for FEA; actual coil winding to be detailed.
6. Hot zone: susceptor wall 15 / base 8; crucible wall 6 / base 7; coil radial gap to insulation bore 15.
7. Geometry from Fusion assembly "All parts of furnace".

## PARTS LIST

| ITEM | DESCRIPTION | MATERIAL | QTY | VOL. [L] |
|---|---|---|---|---|
| 1 | Vessel: shell, top flange, bottom plate | SS304 | 1 | 17.8 |
| 2 | Thermal insulation | Graphite felt | 1 | 54.7 |
| 3 | Induction coil (heater) | Graphite IG-110* | 1 | 1.6 |
| 4 | Susceptor / crucible holder | Graphite IG-110 | 1 | 2.1 |
| 5 | Crucible | Fused silica | 1 | 0.7 |
| 6 | Charge | Si (melt) | – | 2.8 |

* As modelled in FEA. Volumes from 3D model.

REV A 2026-10-18 First issue (ports to follow in Rev B)

COMPANY

**MPrep Engineering**

Material Preparation Engineering

TITLE

**CZ-R4 Crystal-Growth Furnace**

General Arrangement – Section & Hot-Zone Detail

| DRAWING NO. | REV | SHEET |
|---|---|---|
| **MPE-CZR4-GA-001** | **A** | A3 · 1 / 1 |

| SCALE | UNITS | STATUS |
|---|---|---|
| 1:5 (DET. 1:2.5) | mm | **CONCEPT – NOT FOR CONSTR.** |

| DRAWN | DATE |
|---|---|
| Ilham Aksan Maulana | 2026-10-18 |